

Diabetes Risk Analysis

Executive Summary Report — Pima Indians Diabetes Dataset

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Dataset	Pima Indians Diabetes Database (768 patients, 9 clinical variables)
Source	UCI Machine Learning Repository / NIDDK, via GitHub (jbrownlee/Datasets)
Methods	Data cleaning, univariate & bivariate EDA, correlation analysis, hypothesis testing (t-tests, chi-square, ANOVA)
Deliverables	Jupyter Notebook (.ipynb), cleaned CSV, this executive PDF

1. Dataset Overview

The dataset contains diagnostic measurements for 768 female patients of Pima Indian heritage, aged 21 and above, with the goal of predicting diabetes onset (Outcome: 1 = diabetic, 0 = non-diabetic). It includes 8 clinical predictors: Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, and Age.

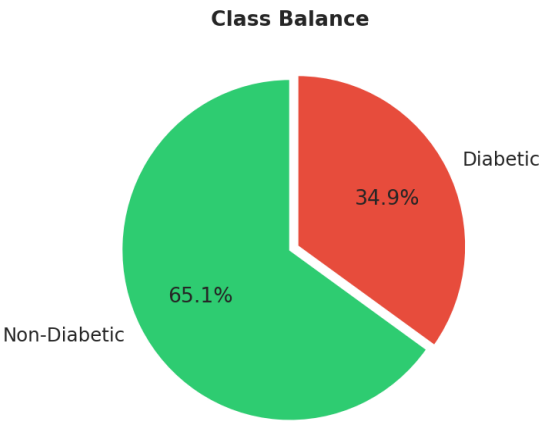


Figure 1: Class balance — 65.1% non-diabetic vs 34.9% diabetic.

2. Data Cleaning Summary

Five clinical columns (Glucose, BloodPressure, SkinThickness, Insulin, BMI) contained physiologically impossible zero values, which were treated as hidden missing data rather than true measurements.

Column	% Missing (as 0)	Imputation Method
Insulin	48.7%	Median, grouped by Outcome
SkinThickness	29.6%	Median, grouped by Outcome
BloodPressure	4.6%	Median, grouped by Outcome
BMI	1.4%	Median, grouped by Outcome
Glucose	0.7%	Median, grouped by Outcome

Outliers were detected using the Tukey IQR rule (1.5x IQR fences) on every numeric feature and were **capped (winsorized)** at the fences rather than removed, preserving all 768 patient records while limiting the influence of extreme values on downstream correlation and hypothesis testing.

3. Feature Correlation Analysis

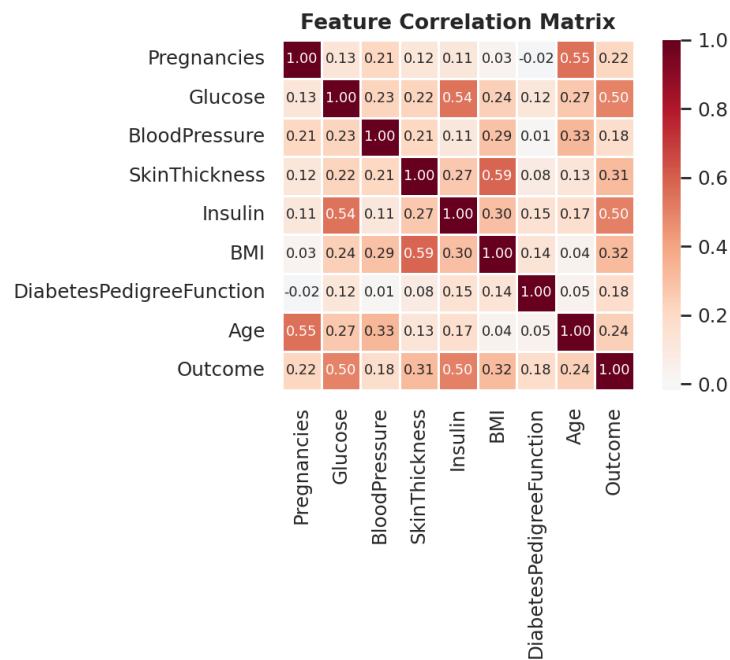


Figure 2: Correlation matrix across all clinical variables.

Glucose (r=0.50) and **Insulin (r=0.50)** show the strongest correlation with diabetes outcome, followed by **BMI (r=0.32)** and **SkinThickness (r=0.31)**. SkinThickness and Insulin also correlate strongly with each other, indicating mild multicollinearity to account for in any predictive model built on this data.

4. Bivariate Patterns by Outcome

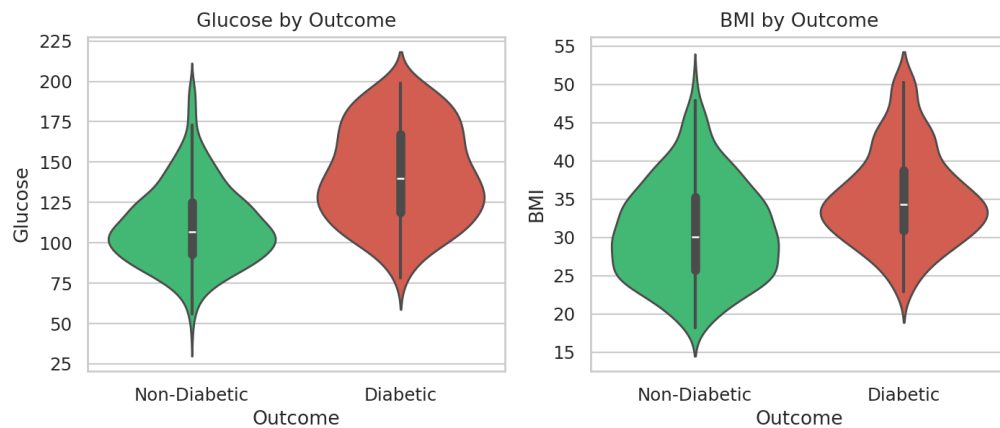


Figure 3: Glucose and BMI distributions for diabetic vs non-diabetic patients.

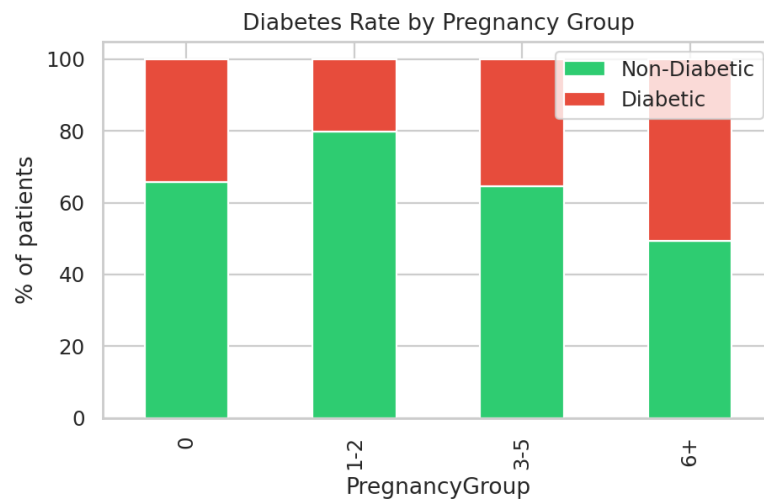


Figure 4: Diabetes prevalence rises with pregnancy count, sharpest at 6+ pregnancies.

5. Statistical Hypothesis Testing

Welch's two-sample t-tests ($\alpha = 0.05$) were run comparing every clinical feature between diabetic and non-diabetic groups.

Feature	Mean (Non-Diabetic)	Mean (Diabetic)	p-value	Effect Size (Cohen's d)
Insulin	111.8	172.8	<0.001	1.21 (large)
Glucose	110.6	142.3	<0.001	1.20 (large)
SkinThickness	27.1	32.1	<0.001	0.69 (medium)
BMI	30.8	35.3	<0.001	0.70 (medium)
Age	31.1	37.1	<0.001	0.52 (medium)
Pregnancies	3.3	4.8	<0.001	0.47 (small-med)

BloodPressure	70.8	75.2	<0.001	0.38 (small)
DiabetesPedigreeFunction	0.4	0.5	<0.001	0.39 (small)

All 8 features are statistically significant ($p < 0.05$), but effect sizes vary widely. **Insulin and Glucose show large practical effects**; BloodPressure and Diabetes Pedigree Function are statistically different but practically weak discriminators.

A chi-square test confirmed a significant association between pregnancy-count category and diabetes outcome ($p < 0.05$). A one-way ANOVA confirmed that mean glucose differs significantly across BMI categories (Underweight/Normal/Overweight/Obese), $p < 0.001$.

6. Actionable Insights & Recommendations

1. **Prioritise Glucose and Insulin testing.** These two features have the strongest correlation with diabetes and the largest effect sizes — they should anchor any screening protocol or predictive model.
2. **Track BMI and Age as secondary risk signals.** Both show significant, moderate effects and rising glucose trends across BMI bands, supporting routine monitoring.
3. **Deprioritise BloodPressure and DiabetesPedigreeFunction as standalone screening variables.** They are statistically different between groups but contribute the least discriminating power of the eight features.
4. **Flag high pregnancy-count patients for diabetes screening.** Diabetes prevalence rises sharply for women with 6+ pregnancies (chi-square $p < 0.05$) — a useful signal for maternal health programs.
5. **Handle missing data carefully in any downstream model.** Nearly half of Insulin values and almost a third of SkinThickness values were originally missing (coded as 0). Group-median imputation, as used here, is preferable to naive zero-filling.
6. **Account for class imbalance (65%/35%).** Any classification model trained on this data should use stratified sampling and evaluate with F1-score or ROC-AUC rather than raw accuracy.
7. **Recommended next step:** build a logistic regression or gradient-boosted tree model using Glucose, BMI, Age, Pregnancies, and DiabetesPedigreeFunction as the primary feature set, given their statistical significance and manageable multicollinearity.

Full analysis, code, and all supporting charts are available in the accompanying Jupyter Notebook (diabetes_analysis.ipynb) and cleaned dataset (diabetes_cleaned.csv).